

Probability (9 th std)

Choose the Appropriate Answer

1. A number between 0 and 1 that is used to measure uncertainty is called
(a) Random variable (b) Trial (c) Simple event (d) Probability
Ans: (d) Probability
2. Probability lies between
(a) -1 and +1 (b) 0 and 1 (c) 0 and n (d) 0 and ∞
Ans: (b) 0 and 1
3. The probability based on the concept of relative frequency theory is called
(a) Empirical probability (b) Classical probability
(c) Both (a) and (b) (d) Neither (a) nor (b)
Ans: (a) Empirical probability
4. The probability of an event cannot be
(a) Equal to zero (b) Greater than zero
(c) Equal to one (d) Less than zero
Ans: (d) Less than zero
5. The probability of all possible outcomes of a random experiment is always equal to
(a) One (b) Zero (c) Infinity (d) Less than one
Ans: (a) One
6. If A is any event in S and its complement is A' then $P(A')$ is equal to
(a) 1 (b) 0 (c) $1 - A$ (d) $1 - P(A)$
Ans: (d) $1 - P(A)$
7. Which of the following cannot be taken as probability of an event?
(a) 0 (b) 0.5 (c) 1 (d) -1
Ans: (d) -1
8. A particular result of an experiment is called
(a) Trial (b) Simple event
(c) Compound event (d) Outcome
Ans: (d) Outcome
9. A collection of one or more outcomes of an experiment is called
(a) Event (b) Outcome (c) Sample point (d) None of these
Ans: (a) Event
10. The six faces of the dice are called equally likely if the dice is
(a) Small (b) Fair (c) Six – faced (d) Round
Ans: (b) Fair

11. Which of the following cannot be taken as probability of an event?

- (a) 0 (b) 0.35 (c) 720 (d) - 720

Ans: (d) -720

12. A letter is chosen at random from the word "MATHEMATICS" the probability of getting a vowel is

- (a) $2/11$ (b) $3/11$ (c) $4/11$ (d) $5/11$

Ans: (c) $4/11$

13. If $P(A) = 1/3$ then $P(A)'$ is

- (a) $1/3$ (b) $2/3$ (c) $3/2$ (d) 1

Ans: (b) $2/3$

14. An integer is chosen from the first twenty natural number, the probability that it is a prime number is

- (a) $1/5$ (b) $1/4$ (c) $3/5$ (d) $4/5$

Ans: (b) $1/4$

15. From a well shuffled pack of 52 cards one card is drawn at random. The probability of getting not a king is

- (a) $12/13$ (b) $1/13$ (c) $4/13$ (d) $2/13$

Ans: (a) $12/13$

16. The probability of each event, when a coin is tossed for 1000 times with frequencies: Head:455 & Tail: 545 is:

- (a) 0.455 & 0.545 (b) 0.5 & 0.5 (c) 0.45 & 0.55 (d) 455 & 545

Ans: (a) 0.455 & 0.545

17. The sum of all probabilities equal to:

- (a) 4 (b) 1 (c) 3 (d) 2

Ans: (b) 1

18. The probability of each event lies between:

- (a) 1 & 2 (b) 1 & 10 (c) 0 & 1 (d) 0 & 5

Ans: (c) 0 & 1

19. If $P(E) = 0.44$, then $P(\text{not } E)$ will be:

- (a) 0.44 (b) 0.55 (c) 0.50 (d) 0.56

Ans: (d) 0.56

$P(E) + P(\text{not } E) = 1 \Rightarrow 0.44 + P(\text{not } E) = 1 \Rightarrow P(\text{not } E) = 1 - 0.44 = 0.56$
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20. If $P(E) = 0.38$, then probability of event E, not occurring is:

- (a) 0.62 (b) 0.38 (c) 0.48 (d) 1

Ans: (a) 0.62

$P(\text{not } E) = 1 - P(E) = 1 - 0.38 = 0.62$

21. The probability of drawing an ace card from a deck of cards is:

- (a) $1/52$ (b) $1/26$ (c) $4/13$ (d) $1/13$

Ans: (d) $1/13$

22. If the probability of an event to happen is 0.3 and the probability of the event not happening is:

- (a) 0.7 (b) 0.6 (c) 0.5 (d) None of the above

Ans: (a) 0.7

$$P(\text{not } E) = 1 - 0.3 = 0.7$$

23. A dice is thrown. The probability of getting 1 and 5 is:

- (a) $\frac{1}{6}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) $\frac{1}{2}$

Ans: (c) $\frac{1}{3}$

24. A batsman hits boundaries for 6 times out of 30 balls. Find the probability that he did not hit the boundaries.

- (a) $\frac{1}{5}$ (b) $\frac{2}{5}$ (c) $\frac{3}{5}$ (d) $\frac{4}{5}$

Ans: (d) $\frac{4}{5}$

$$\begin{aligned} \text{No. of balls without boundaries} &= 30 - 6 = 24 \\ \text{Probability of no boundary} &= 24 / 30 = \frac{4}{5} \end{aligned}$$

25. Three coins were tossed 200 times. The number of times 2 heads came up is 72. Then the probability of 2 heads coming up is:

- (a) $1 / 25$ (b) $2 / 25$ (c) $7 / 25$ (d) $9 / 25$

Ans: (d) $9 / 25$

$$\text{Probability} = 72 / 200 = 9 / 25$$

26. What is the probability of getting an odd number less than 4, if a die is thrown?

- (a) $1 / 6$ (b) $1 / 2$ (c) $1 / 3$ (d) 0

Ans: (c) $1 / 3$

$$\begin{aligned} \text{Sample space, } S &= \{1, 2, 3, 4, 5, 6\} \\ \text{Favourable outcomes} &= \{1, 3\} \\ \text{The probability of getting an odd number less than 4} &= 2 / 6 = \frac{1}{3}. \end{aligned}$$

27. What is the probability of impossible events?

- (a) 1 (b) 0 (c) More than 1 (d) Less than 1

Ans: (b) 0

28. Performing an event once is called

- (a) Sample (b) Trial (c) Error (d) None of the above

Ans: (b) Trial

29. A card is drawn from a well-shuffled deck of 52 cards. What is the probability of getting a king of the red suits?

- (a) $3 / 36$ (b) $1 / 26$ (c) $3 / 26$ (d) $1 / 16$

Ans: (b) $1 / 26$

$$\text{The probability of getting a king of red suits} = 2 / 52 = 1 / 26$$

30. Find the probability of a selected number is a multiple of 4 from the numbers 1, 2, 3, 4, 5 ...15.

- (a) $1/5$ (b) $1/3$ (c) $4/12$ (d) $2/15$

Ans: (a) $1/5$

$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$
Multiples of 4 from the sample space = $\{4, 8, 12\}$
The probability of the selected number is a multiple of 5 is $3/15 = 1/5$

31. What is the probability of drawing a queen from the deck of 52 cards?

- (a) $1/26$ (b) $1/52$ (c) $1/13$ (d) $3/52$

Ans: (c) $1/13$

The probability of drawing a queen from a deck of 52 cards = $4/52 = 1/13$

32. Which of the following cannot be the probability of an event?

- (a) 1 (b) 0 (c) 0.75 (d) 1.3

Ans: (d) 1.3

33. There are 4 green and 2 red balls in a basket. What is the probability of getting the red balls?

- (a) $1/2$ (b) $1/3$ (c) $1/5$ (d) $1/6$

Ans: (b) $1/3$

Total balls = 4 green + 2 red = 6 balls No. of red balls = 2.
The probability of getting the red balls = $2/6 = 1/3$

34. Empirical probability is also known as

- (a) Classic probability (b) Subjective probability
(c) Experimental probability (d) None of the above

Ans: (c) Experimental probability

35. If two coins are tossed simultaneously, then what is the probability of getting exactly two tails?

- (a) $1/4$ (b) $1/2$ (c) $1/3$ (d) None of the above

Ans: (a) $1/4$

$S = \{HH, HT, TH, TT\}$ Favourable outcome = $\{TT\}$
The probability of getting exactly two tails = $1/4$

36. If $P(E) = 0.37$, then $P(\text{not } E)$ will be

- (a) 0.37 (b) 0.63 (c) 0.57 (d) None of these

Ans: (b) 0.63

37. Probability of getting even number in a single throw of dice is

- (a) $\frac{1}{2}$ (b) $\frac{1}{6}$
(c) $\frac{5}{6}$ (d) None of these

Ans: (a) $1/2$

38. There are 5 prizes on 1000 tickets of a lottery of company. Probability of winning prize is

(a) $\frac{199}{200}$

(b) $\frac{1}{200}$

(c) $\frac{198}{200}$

(d) None of these

Ans: (b) 1 / 200

39. Probability of drawing an ace from a deck of 52 cards is

(a) $\frac{1}{52}$

(b) $\frac{1}{26}$

(c) $\frac{1}{13}$

(d) $\frac{3}{52}$

Ans: (c) 1 / 13

40. There are 5 red and 3 black balls in a bag. Probability of drawing a black ball is

(a) $\frac{5}{8}$

(b) $\frac{1}{2}$

(c) $\frac{3}{8}$

(d) None of these

Ans: (c) 3 / 8

41. If the probability of winning a game is 0.3, then probability of losing it is

(a) 0.6

(b) 0.7

(c) 0.5

(d) None of these

Ans: (b) 0.7

42. Probability of getting 10 of a black suit from a deck of 52 cards is

(a) $\frac{1}{52}$

(b) $\frac{1}{26}$

(c) $\frac{1}{13}$

(d) None of these

Ans: (b) 1 / 26

43. A dice is thrown once. Probability of getting a number 3 or 4 is

(a) $\frac{1}{6}$

(b) $\frac{2}{3}$

(c) $\frac{1}{2}$

(d) $\frac{1}{3}$

Ans: (d) 1 / 3

44. There are 50 tickets numbered from 1 to 50 in a box. Probability of drawing a ticket bearing prime number is

(a) $\frac{13}{50}$

(b) $\frac{3}{10}$

(c) $\frac{17}{50}$

(d) None of these

Ans: (b) 3 / 10

45. Probability of a leap year having 53 Sundays is

(a) $\frac{1}{7}$

(b) $\frac{3}{7}$

(c) $\frac{2}{7}$

(d) None of these

Ans: (c) $\frac{2}{7}$

46. Two coins were tossed 200 times and the following results were obtained.

Two heads: 55, One head and one tail: 105, Two tails: 40

What is the probability of event of obtaining minimum one head?

(a) 0.5

(b) 0.8

(c) 0.55

(d) 0.16

Ans: (b) 0.8

Number of events of obtaining minimum one head = $55 + 105 = 160$

$$P(E) = \frac{\text{Number of events of obtaining minimum one head}}{\text{Number of total trials}} = \frac{160}{200} = 0.8.$$

47. A dice was thrown 500 times. Frequencies for the outcomes 1, 2, 3, 4, 5, and 6 are given in the table. What is the probability of getting '4' as outcome?

Outcome	1	2	3	4	5	6
Frequency	78	80	93	79	91	79

(a) 0.16

(b) 0.158

(c) 0.156

(d) 0.131

Ans: (d) 0.131

$$\text{Probability of getting '4' as outcome} = \frac{\text{Event of occurrence of getting '4'}}{\text{Total number of trials}} = 0.158 = \frac{79}{500}$$

48. Marks obtained by a student in a test is shown in the table below. What is the probability that the student has scored more than 80?

Test no.	1	2	3	4	5
Marks	81	87	76	70	90

(a) 0.6

(b) 0.8

(c) 0.4

(d) 0.5

Ans: (a) 0.6

$$\text{Probability that the student has scored more than 80} = \frac{\text{number of occurrence of event}}{\text{Total number of trials}} = \frac{3}{5} = 0.6.$$

49. 1000 families with 2 children were studied and the following data was collected.
What is the probability that the family has at least one boy?

Number of boys in the family	0	1	2
Number of families	270	415	315

- (a) 0.415 (b) 0.270 (c) 0.73 (d) 0.315

Ans: (c) 0.73

$$\text{The probability that the family has at least one boy} = \frac{\text{number of families having at least one boy}}{\text{Total number of families}} = \frac{730}{1000} = 0.73.$$

- 1500 drivers were selected for a study to find a relationship between age and accidents. The data is shown in the table below. Answer the questions (50 - 52).

Age of drivers(in years)	Number of accidents		
	0	1	More than 1
18-35	285	325	90
35-50	145	277	78
Above 50	123	118	59

50. What is the probability of being 18 - 35 years of age and having more than 1 accidents?

- (a) 0.06 (b) 0.6 (c) 0.08 (d) 0.1

Ans: (a) 0.06

$$\text{Probability of being 18-35 years of age and having more than 1 accidents} = \frac{90}{1500} = 0.06$$

51. What is the probability of being elder than 35 years of age and having at least one accident?

- (a) 0.41 (b) 0.25 (c) 0.354 (d) 0.333

Ans: (c) 0.354

$$\begin{aligned} \text{Elder than 35 years of age and having at least 1 accident} &= 277 + 78 + 118 + 59 = 532 \\ \text{Total number of events} &= 1500 \\ \text{Probability of being elder than 35 years of age and having at least 1 accident} &= \frac{532}{1500} = 0.354. \end{aligned}$$

52. What is the probability of being 35 - 50 years of age and having no accidents?

- (a) 0.29 (b) 0.35 (c) 0.09 (d) 0.08

Ans: (c) 0.09

$$\text{35-50 years of age and having no accidents} = \frac{145}{1500} = 0.09$$

53. There are 6 marbles in a box with number 1 to 6 marked on each of them. What is the probability of drawing a marble with number 2?

(a) $\frac{1}{6}$

(b) $\frac{1}{5}$

(c) $\frac{1}{3}$

(d) 1

Ans: (a) $\frac{1}{6}$

54. A coin is flipped to decide which team starts the game. What is the probability of your team will start?

(a) $\frac{1}{4}$

(b) $\frac{1}{2}$

(c) 1

(d) 0

Ans: (b) $\frac{1}{2}$

55. A die is thrown once. What will be the probability of getting a prime number?

(a) $\frac{1}{6}$

(b) $\frac{1}{2}$

(c) 1

(d) 0

Ans: (b) $\frac{1}{2}$

Cards marked with numbers 1 to 25 are placed in the box and mixed thoroughly. One card is drawn at random from the box. Answer the following questions (Q 56 – Q 65).

56. What is the probability of getting a number 5?

(a) 1

(b) 0

(c) $\frac{1}{25}$

(d) $\frac{1}{5}$

Ans: (a) $\frac{1}{25}$

57. What is the probability of getting a number less than 11?

(a) 1

(b) 0

(c) $\frac{1}{5}$

(d) $\frac{2}{5}$

Ans: (d) $\frac{2}{5}$

58. What is the probability of getting a number greater than 25?

(a) 1

(b) 0

(c) $\frac{1}{5}$

(d) $\frac{2}{5}$

Ans: (b) 0

59. What is the probability of getting a multiple of 5?

(a) 1

(b) 0

(c) $\frac{1}{5}$

(d) $\frac{2}{5}$

Ans: (c) $\frac{1}{5}$

60. What is the probability of getting an even number?

(a) $\frac{8}{25}$

(b) $\frac{9}{25}$

(c) $\frac{12}{25}$

(d) $\frac{13}{25}$

Ans: (c) $\frac{12}{25}$

61. What is the probability of getting an odd number?

(a) $\frac{8}{25}$

(b) $\frac{9}{25}$

(c) $\frac{12}{25}$

(d) $\frac{13}{25}$

Ans: (d) $\frac{13}{25}$

62. What is the probability of getting a prime number?

(a) $\frac{8}{25}$

(b) $\frac{9}{25}$

(c) $\frac{12}{25}$

(d) $\frac{13}{25}$

Ans: (a) $\frac{8}{25}$

63. What is the probability of getting a number divisible by 3?

(a) $\frac{8}{25}$

(b) $\frac{9}{25}$

(c) $\frac{6}{25}$

(d) $\frac{13}{25}$

Ans: (c) $\frac{6}{25}$

64. What is the probability of getting a number divisible by 4?

(a) $\frac{8}{25}$

(b) $\frac{9}{25}$

(c) $\frac{6}{25}$

(d) $\frac{3}{25}$

Ans: (d) $\frac{3}{25}$

65. What is the probability of getting a number divisible by 7?

(a) $\frac{8}{25}$

(b) $\frac{9}{25}$

(c) $\frac{6}{25}$

(d) $\frac{3}{25}$

Ans: (d) $\frac{3}{25}$

66. A bag has 4 red balls and 2 yellow balls. A ball is drawn from the bag without looking into the bag. What is probability of getting a red ball?

(a) $\frac{1}{6}$

(b) $\frac{2}{3}$

(c) $\frac{1}{3}$

(d) 1

Ans: (b) $\frac{2}{3}$

67. A bag has 4 red balls and 2 yellow balls. A ball is drawn from the bag without looking into the bag. What is probability of getting a yellow ball?

(a) $\frac{1}{6}$

(b) $\frac{2}{3}$

(c) $\frac{1}{3}$

(d) 1

Ans: (c) $\frac{1}{3}$

A box contains 3 blue, 2 white, and 5 red marbles. If a marble is drawn at random from the box, then answer the questions from 68 to 72.

68. What is the probability that the marble will be white?

(a) $\frac{1}{6}$

(b) $\frac{1}{5}$

(c) $\frac{1}{3}$

(d) 1

Ans: (b) $\frac{1}{5}$

69. What is the probability that the marble will be red?

(a) $\frac{1}{5}$

(b) $\frac{1}{2}$

(c) 1

(d) 0

Ans: (b) $\frac{1}{5}$

70. What is the probability that the marble will be blue?

(a) $\frac{3}{10}$

(b) $\frac{1}{2}$

(c) 1

(d) 0

Ans: (a) $\frac{3}{10}$

71.What is the probability that the marble will be any one color?

(a) $\frac{1}{6}$

(b) $\frac{1}{2}$

(c) 1

(d) 0

Ans: (c) 1

72.What is the probability that the marble will be red or blue?

(a) 1

(b) $\frac{4}{5}$

(c) $\frac{1}{5}$

(d) $\frac{2}{5}$

Ans: (b) 4 / 5

A die is thrown once, then answer the questions from 73 to 77.

73.Find the probability of getting a prime number.

(a) $\frac{1}{6}$

(b) $\frac{1}{2}$

(c) 1

(d) 0

Ans: (b) 1 / 2

74.Find the probability of getting a number lying between 2 and 6

(a) $\frac{1}{6}$

(b) $\frac{1}{2}$

(c) 1

(d) 0

Ans: (b) 1 / 2

75.Find the probability of getting an odd number.

(a) $\frac{1}{6}$

(b) $\frac{1}{2}$

(c) 1

(d) 0

Ans: (b) 1 / 2

76.Find the probability of getting an even number.

(a) $\frac{1}{6}$

(b) $\frac{1}{2}$

(c) 1

(d) 0

Ans: (b) 1 / 2

77.Find the probability of getting a number greater than 4.

(a) $\frac{1}{6}$

(b) $\frac{2}{3}$

(c) $\frac{1}{3}$

(d) 1

Ans: (c) 1 / 3

A box contains 5 red marbles, 6 white marbles and 4 green marbles. If a marble is drawn at random from the box, then answer the questions from 78 to 83.

78.What is the probability that the marble will be white?

(a) $\frac{1}{6}$

(b) $\frac{2}{5}$

(c) $\frac{1}{3}$

(d) 1

Ans: (b) 2 / 5

79.What is the probability that the marble will be red?

(a) $\frac{1}{6}$

(b) $\frac{2}{3}$

(c) $\frac{1}{3}$

(d) 1

Ans: (c) 1 / 3

80. What is the probability that the marble will be green?

- (a) 0.3 (b) $\frac{1}{2}$ (c) 1 (d) none of these

Ans: (d) None of these

81. What is the probability that the marble will be any one color?

- (a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) 1 (d) 0

Ans: (c) 1

82. What is the probability that the marble will be red or green?

- (a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) 1 (d) None of these

Ans: (d) None of these

83. What is the probability that the marble will be blue?

- (a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) 1 (d) 0

Ans: (d) 0

Cards marked with numbers 1 to 50 are placed in the box and mixed thoroughly. One card is drawn at random from the box. Answer the following questions from (84 – 93).

84. What is the probability of getting a number 5?

- (a) 1 (b) 0 (c) $\frac{1}{25}$ (d) $\frac{1}{50}$

Ans: (d) 1 / 50

85. What is the probability of getting a number less than 11?

- (a) 1 (b) 0 (c) $\frac{1}{5}$ (d) $\frac{2}{5}$

Ans: (c) 1 / 5

86. What is the probability of getting a number greater than 50?

- (a) 1 (b) 0 (c) $\frac{1}{5}$ (d) $\frac{2}{5}$

Ans: (b) 0

87. What is the probability of getting a multiple of 5?

- (a) 1 (b) 0 (c) $\frac{1}{25}$ (d) $\frac{1}{5}$

Ans: (d) 1 / 5

88. What is the probability of getting an even number?

- (a) 1 (b) $\frac{1}{2}$ (c) $\frac{12}{25}$ (d) $\frac{13}{25}$

Ans: (c) 12 / 25

89.What is the probability of getting an odd number?

- (a) 1 (b) $\frac{1}{2}$ (c) $\frac{12}{25}$ (d) $\frac{13}{25}$

Ans: (d) 13 / 25

90.What is the probability of getting a prime number?

- (a) 1 (b) $\frac{1}{2}$ (c) $\frac{4}{10}$ (d) $\frac{3}{10}$

Ans: (c) 4 / 10

91.What is the probability of getting a number divisible by 3?

- (a) $\frac{8}{25}$ (b) $\frac{9}{25}$ (c) $\frac{12}{25}$ (d) $\frac{13}{25}$

Ans: (a) 8 / 25

92.What is the probability of getting a number divisible by 4?

- (a) $\frac{8}{25}$ (b) $\frac{9}{25}$ (c) $\frac{6}{25}$ (d) $\frac{3}{25}$

Ans: (c) 6 / 25

93.What is the probability of getting a number divisible by 7?

- (a) $\frac{8}{25}$ (b) $\frac{9}{25}$ (c) $\frac{6}{25}$ (d) $\frac{3}{25}$

Ans: (d) 3 / 25

94.Find the probability that any of these bags chosen at random contains more than 5 kg of flour.

- (a) 1 / 12 (b) 7 / 12 (c) 3 / 4 (d) None of these

Ans: (b) 7 / 12

A company selected 4000 households at random and surveyed them to find out a relationship between income level and the number of mobile sets in a home. The information so obtained is listed in the following table. Answer the following questions from (95 – 99).

Monthly income	No of Mobile sets			
	0	1	2	Above2
< 10000	20	80	10	.
10000 - 14999	10	240	60	.
15000 - 19999	.	380	120	30
20000 - 24999	.	520	370	80
25000 and above	.	1100	760	220

95. Find the probability of a household earning Rs.10000 — Rs.14999 per year and having exactly one mobile set

- (a) 0.6 (b) 0.8 (c) 0.4 (d) None of these

Ans: (a) 0.6

$$\text{Probability} = 240 / 4000 = 0.6$$

96. In the above Question, find the probability of a household earning more than 25000 per year and having exactly 2 mobile set

- (a) 2 (b) 19 (c) 12 (d) 3

Ans: (b) 19

$$\text{Probability} = 760 / 4000 = 19$$

97. In the above Question, find the probability of a household earning more than 25000 per year and having 2 or more mobile set

- (a) 245 (b) 3 (c) 1 (d) None of these

Ans: (a) 245

$$\text{Probability} = 980 / 4000 = 245$$

98. In the above Question, find the probability of a household having no mobile set at all?

- (a) 3 / 400 (b) 1 / 400 (c) 1 / 200 (d) None of these

Ans: (a) 3 / 400

$$\text{Probability} = 30 / 4000 = 3 / 400$$

99. In the above Question, find the probability of a household having 3 mobile set and having income less than 10000

- (a) 1 (b) 0 (c) 24 (d) None of these

Ans: (b) 0

100. The probability of the events lies between

- (a) $-1 \leq p \leq 1$ (b) $0 \leq p \leq 1$ (c) $-1 \leq p \leq 0$ (d) $-1 < p \leq 1$

Ans: (b) $0 \leq p \leq 1$